

Chapter 4: Effect Modification

Applied Use Case: Did the Retention Offer Actually Reduce Churn?

1 Heterogeneity of Treatment Effects (Section 4.1, p. 43–45)

In Chapters 1–3, our focus was estimating the **Average Causal Effect (ACE)** across the entire population ($E[Y^{a=1} - Y^{a=0}]$). However, a single overall average can mask vital business insights:

- **Subgroup Variations:** An intervention rarely affects every customer identically.
- **Net Zero Masking:** Suppose sending a retention offer ($A = 1$) reduces churn by 15% among highly engaged subscribers, but **increases** churn by 15% among inactive subscribers (*Sleeping Dogs*). If each group constitutes 50% of the database, the overall average causal effect is exactly **zero**:

$$E[Y^{a=1} - Y^{a=0}] = 0.5(-0.15) + 0.5(+0.15) = 0$$

An unstratified analysis would incorrectly conclude that the retention offer is ineffective, whereas it is highly effective when targeted appropriately.

2 Definition of an Effect Modifier (V) (Section 4.1, p. 44–45)

A baseline variable V (e.g., tenure, engagement score, customer segment) is an **effect modifier** of intervention A on outcome Y if the causal effect of A on Y varies across levels/strata of V .

2.1 Scale Dependence: Additive vs. Multiplicative

Effect modification depends strictly on the measure of causal effect being analyzed:

1. Additive Scale (Causal Risk Difference):

$$E[Y^{a=1} - Y^{a=0} \mid V = 1] \neq E[Y^{a=1} - Y^{a=0} \mid V = 0]$$

2. Multiplicative Scale (Causal Risk Ratio):

$$\frac{\Pr[Y^{a=1} = 1 \mid V = 1]}{\Pr[Y^{a=0} = 1 \mid V = 1]} \neq \frac{\Pr[Y^{a=1} = 1 \mid V = 0]}{\Pr[Y^{a=0} = 1 \mid V = 0]}$$

Qualitative vs. Quantitative Effect Modification

- **Qualitative Effect Modification:** The intervention helps in one stratum ($RD < 0$) but harms in another ($RD > 0$). This holds on **both** additive and multiplicative scales.
- **Quantitative Effect Modification:** The intervention acts in the same direction across all strata, but with differing magnitude.

3 Stratification to Identify Effect Modification (Section 4.2, p. 45–47)

To detect effect modification by V , we partition the data into strata defined by V and estimate stratum-specific causal effects. If treatment assignment is observational, we adjust for baseline confounders L within each stratum using Standardization or IP Weighting:

$$\Pr[Y^a = 1 \mid V = v] = \sum_l \Pr[Y = 1 \mid A = a, V = v, L = l] \cdot \Pr[L = l \mid V = v]$$

3.1 Causal vs. Surrogate Effect Modifiers (Section 4.2, p. 46–47; Ch. 6.6, p. 84)

- **Causal Effect Modifier:** The variable V directly alters the mechanism through which treatment A impacts outcome Y (e.g., price sensitivity directly modifies discount responsiveness).
- **Surrogate Effect Modifier:** V itself does not alter the mechanism, but is correlated with an unmeasured true causal modifier (e.g., *geographic region* proxying for local purchasing power).

4 Practical Business Significance (Section 4.3, p. 47–49)

1. **Targeting & ROI Optimization:** For decision-makers allocating marketing budget, the **additive scale (Risk Difference)** is paramount. It determines the Number Needed to Treat ($\text{NNT} = 1/|\text{RD}|$) per segment, identifying exactly where offers yield positive net revenue.
2. **Transportability / External Validity (Fine Point 4.2, p. 50):** An A/B test conducted in Market A (e.g., Germany) yielding a 5% churn reduction cannot automatically be generalized to Market B (e.g., Poland) if the distribution of effect modifier V differs between markets. Stratum-specific effects ($\Pr[Y^a = 1 \mid V = v]$) are far more **transportable** across populations than overall average effects.
3. **Mechanistic Insights:** Identifying which customer segments respond to discounts helps product teams refine customer onboarding and product features.

5 Confounding Control vs. Effect Modification (Section 4.4–4.6, p. 49–54)

Key Methodological Distinction

- **Confounders (L):** Nuisance variables causing bias. We want to adjust for them and synthesize/marginalize them out.
- **Effect Modifiers (V):** Variables of primary scientific/business interest. We want to report separate stratum-specific estimates rather than average them away.

5.1 Pitfalls of Pure Stratification for Confounding Control

Using stratification as a primary method to control for a large set of confounders L suffers from:

1. **Curse of Dimensionality (Section 4.4, p. 50):** As the number of covariates increases, cell sizes rapidly shrink to zero, causing positivity violations.
2. **Noncollapsibility of Odds Ratios (Fine Point 4.3, p. 54):** Conditional odds ratios across strata do not average to the marginal odds ratio, even in the complete absence of confounding.

Conclusion: Use **IP Weighting** or **Standardization** for confounding adjustment, and reserve **Stratification** for evaluating effect modification across key target covariates V .